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Attribution of Pacific trend discrepancies using the Forced Component Estimation Statistical Method Intercomparison Project (ForceSMIP)

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1 Introduction and objectives

The pattern of Pacific sea-surface temperature (SST) change since 1980 has been highlighted as a key inconsistency between climate models and observations, with widespread impacts on the hydrological cycle, hurricane activity, sea level rise, and climate sensitivity. However, it is unknown whether this trend discrepancy results from discrepancies in the forced warming pattern simulated by climate models or discrepancies in simulated internal variability. Here, we use output of the Forced Component Estimation Statistical Method Intercomparison Project (ForceSMIP) to assess the forced and unforced contributions to Pacific SST trend discrepancies.

2 Methods and data

ForceSMIP Tier1 (Wills et al. 2026) evaluated 30 existing and newly developed methods for estimating the spatiotemporally varying forced response in key fields such as SST and precipitation from single realizations of the climate system. These methods were tested or trained by community contributors using five climate model large ensembles, evaluated using a standardized set of benchmark ensemble members taken from various climate models, and applied to observations. We quantify their skill within the evaluation members based on root mean square error and pattern correlation in long-term trends and compare their skill to several reference methods. We assess the bias-variance tradeoff within ForceSMIP Tier1 based on the pattern correlation between each method's estimates of the forced response in different evaluation members.

3 Preliminary results

We highlight a bias-variance tradeoff amongst the statistical and machine learning methods that show skill in forced response estimation, where methods that reduce the variance in estimated trends the most exhibit biases learned from the climate-model-based training data. Low-bias high-variance methods indicate a large forced contribution to the model-observations SST-trend discrepancy, whereas low-variance high-bias methods assess the SST-trend discrepancy to be mostly due to internal variability. The latter category of methods relies on training data from climate models with documented systematic biases, and we therefore suggest that more weight be put into the former category of methods, which would lead to the conclusion that there is a large forced contribution to the model-observations SST-trend discrepancy. Our work illustrates the value of statistical attribution methods that are not reliant on climate models for interpreting trend discrepancies between climate models and observations. It also motivates ForceSMIP Tiers 2 and 3, which has been specifically designed with more evaluation ensemble members to assess the bias-variance tradeoff more rigorously.

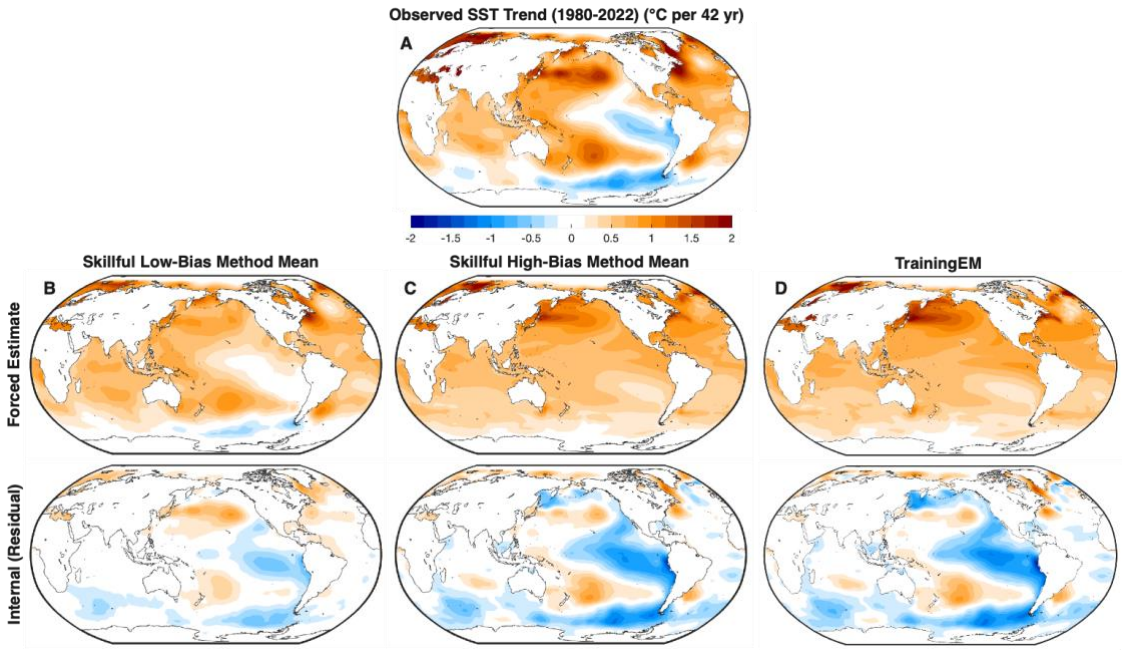

Figure 1. (A) Observed SST trends over 1980-2022 and the forced and internal components assessed by (B) skillful low-bias high-variance methods, (C) skillful low-variance high-bias methods, and (D) the scaled ensemble mean of the training models. Methods are assessed as low-variance high-bias or low-bias high-variance based on the high or low pattern correlation between their estimates of the forced response in different evaluation members, respectively. All data is from ForceSMIP Tier 1 (Wills et al. 2026).

References

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